

01.1 — Schema Reference

Audience: Developers + technical admins working with Haritha's ERPNext customization layer. **Goal:** Catalog every customization (Custom Field, Property Setter, Print Format, Notification, Letter Head) captured in the `haritha_hospital` custom app. **Last updated:** 2026-08-29 **Source data:** `apps/haritha_hospital/haritha_hospital/fixtures` (78 CF + 189 PS + 5 PF + 8 N + 2 LH = 282 rows; **Haritha-tagged subset** = 78 + 189 + 5 + 6 + 2 = 280; see "Note on counts" below)

Overview

Haritha Hospitals uses 4 Frappe apps: - **frappe** v16.30.0 — framework - **erpnext** v16.30.0 — accounting/inventory/sales/buying - **hrms** v16.5.0 — HR/payroll/shift/attendance (pinned per LEARNINGS #44) - **payments** — gateway integrations

In addition, the **haritha_hospital** custom app captures all customizations as fixtures. **282 rows** are exported as fixtures in total, broken down by DocType:

DocType	Total rows in fixture	Notes
Custom Field	78	All 78 are Haritha-tagged (module not back-filled on Custom Fields by Frappe — see LEARNINGS #152)
Property Setter	189	All 189 are Haritha-tagged (same reason)
Print Format	48	2 module=HR + 3 module=NULL (old stock seeds — see LEARNINGS #156) + 43 module=stock
Notification	8	3 module=HR + 3 module=stock + 2 module=NULL
Letter Head	2	All 2 (Letter Head has no module column — see LEARNINGS #155)

Note on counts: the GUIDE and TRACKER claim **274 = 78 + 189 + 3 + 2 + 2**. The fixture export (the ground truth) actually returns **282 = 78 + 189 + 48 + 8 + 2**, of which the Haritha-tagged subset is **280 = 78 + 189 + 5 + 6 + 2**. The discrepancy comes from stock seed rows (old Print Formats / Notifications with `module=NULL`) being included in the export. For the purpose of this doc, the **fixture file counts are authoritative** — they are what bench migrate actually applies. The discrepancy is documented here so future audits don't chase a phantom.

How to read this doc

Column	Meaning
DocType	The Frappe/ERPNext/HRMS table the customization is on
Field / Row	Identifier — fieldname for CFs/PSs, name for PFs/Ns/LHs
Type	Fieldtype for CFs (Data, Link, Select, Check, ...); property for PSs (hidden, reqd, default, options, ...); blank for PF/N/LH
Reqd	Whether the field is mandatory (Custom Field reqd property)
Value	Property Setter value / Print Format disabled / Notification enabled
Purpose	Inferred from name + Frappe framework knowledge

Abbreviations: - **CF** — Custom Field - **PS** — Property Setter - **PF** — Print Format - **N** — Notification - **LH** — Letter Head - **reqd** — required (mandatory) - **HRMS** — Frappe HRMS app - **ERPNext** — Frappe ERPNext app - **Haritha** — Haritha Hospitals custom app (haritha_hospital)

Custom Fields by DocType

78 Custom Fields across **23 DocTypes**, grouped below.

Employee (HRMS) — 20 Custom Fields

Field	Label	Type	Reqd	Purpose
employment_type	Employment Type	Link → Employment Type	No	Employee's contract type (Full-time, Part-time, Contract, etc.)
job_applicant	Job Applicant	Link → Job Applicant	No	Back-link to the Job Applicant record that became this Employee
grade	Grade	Link → Employee Grade	No	Pay grade / seniority level
default_shift	Default Shift	Link → Shift Type	No	Shift used by Auto Attendance cron when no Shift Assignment exists
health_insurance_section	Health Insurance	Section Break	No	Layout separator for the health insurance block

Field	Label	Type	Reqd	Purpose
health_insurance_health_insurance_provider	Health Insurance Provider	Link → Employee Health Insurance	No	Insurer name
health_insurance_health_insurance_no	Health Insurance No	Data	No	Policy number
approvers_section	Approvers	Section Break	No	Layout separator for the approvers block
expense_approver	Expense Approver	Link → User	No	User who approves this employee's Expense Claims
leave_approver	Leave Approver	Link → User	No	User who approves this employee's Leave Applications
column_break_45	(column break)	Column Break	No	Layout — column break in the approvers row
shift_request_approver	Shift Request Approver	Link → User	No	User who approves this employee's Shift Requests
employee_advance_employee_advance_account	Employee Advance Account	Link → Account	No	GL account for posting Employee Advances against this employee
salary_cb	(column break)	Column Break	No	Layout — column break in the bank row
payroll_cost_center	Payroll Cost Center	Link → Cost Center	No	Cost Center that this employee's payroll gets allocated to
bank_cb	(column break)	Column Break	No	Layout — column break in the bank row
ifsc_code	IFSC Code	Data	No	Bank branch IFSC for NEFT/IMPS payroll
pan_number	PAN Number	Data	No	Permanent Account Number for TDS calculation
micr_code	MICR Code	Data	No	Bank branch MICR for cheque printing

Field	Label	Type	Reqd	Purpose
provident_fund_account	Provident Fund Account	Data	No	PF account number (UAN linked)

Company (ERPNext) — 11 Custom Fields

Field	Label	Type	Reqd	Purpose
hr_and_payroll_tab	HR & Payroll	Tab Break	No	Adds a new “HR & Payroll” tab to Company form
hr_settings_section	HR & Payroll Settings	Section Break	No	Layout separator
default_expense_claim_payable_account	Default Expense Claim Payable Account	Link → Account	No	GL account for pending Expense Claim reimbursements
default_employee_advance_account	Default Employee Advance Account	Link → Account	No	GL account for Employee Advance postings
column_break_10	(column break)	Column Break	No	Layout
default_payroll_payable_account	Default Payroll Payable Account	Link → Account	No	GL account for net Salary payable
hra_section	HRA Settings	Section Break	No	Layout separator for HRA configuration
basic_component	Basic Component	Link → Salary Component	No	The “Basic” salary component used in structures
hra_component	HRA Component	Link → Salary Component	No	The “House Rent Allowance” component
hra_column_break	(column break)	Column Break	No	Layout
arrear_component	Arrear Component	Link → Salary Component	No	Component for back-dated salary adjustments

Employee Tax Exemption Proof Submission (HRMS) — 9 Custom Fields

Field	Label	Type	Reqd	Purpose
hra_section	HRA Exemption	Section Break	No	Layout separator

Field	Label	Type	Reqd	Purpose
house_rent_payment	House Rent Payment Amount	Currency	No	Total rent paid in the financial year
rented_in_metro	Rented in Metro City	Check	No	Flag for HRA exemption calculation (metro = 50% of basic, non-metro = 40%)
rented_from_date	Rented From Date	Date	No	Tenancy start date
rented_to_date	Rented To Date	Date	No	Tenancy end date
hra_column_break	(column break)	Column Break	No	Layout
monthly_house_rent	Monthly House Rent	Currency	No	Per-month rent (auto-derivable from annual / 12)
monthly_hra_exemption	Monthly Eligible Amount	Currency	No	Per-month HRA-exempt amount (computed)
total_eligible_hra_exemption	Total Eligible HRA Exemption	Currency	No	Total HRA exemption claim for the year

Department (HRMS) — 8 Custom Fields

Field	Label	Type	Reqd	Purpose
section_break_4	(section break)	Section Break	No	Layout separator
payroll_cost_center	Payroll Cost Center	Link → Cost Center	No	Cost Center allocation for this department's payroll
column_break_9	(column break)	Column Break	No	Layout
leave_block_list	Leave Block List	Link → Leave Block List	No	Department-level leave block (e.g., "no leave during Dec 31 inventory")
approvers	Approvers	Section Break	No	Layout separator
shift_request_approver	Shift Request Approver	Table → Department Approver	No	Multi-row approver list for Shift Requests routed by department

Field	Label	Type	Reqd	Purpose
leave_approvers	Leave Approver	Table → Department Approver	No	Multi-row approver list for Leave Applications routed by department
expense_approver	Expense Approver	Table → Department Approver	No	Multi-row approver list for Expense Claims routed by department

Employee Tax Exemption Declaration (HRMS) — 7 Custom Fields

Field	Label	Type	Reqd	Purpose
hra_section	HRA Exemption	Section Break	No	Layout separator
monthly_house_rent	Monthly House Rent	Currency	No	Expected per-month rent
rented_in_metro_city	Rented in Metro City	Check	No	Metro flag for the declaration
salary_structure_hra	HRA as per Salary Structure	Currency	No	HRA component value per salary structure
hra_column_break	(column break)	Column Break	No	Layout
annual_hra_exemption	Annual HRA Exemption	Currency	No	Projected annual exemption
monthly_hra_exemption	Monthly HRA Exemption	Currency	No	Per-month projection

Designation (HRMS) — 3 Custom Fields

Field	Label	Type	Reqd	Purpose
appraisal_template	Appraisal Template	Link → Appraisal Template	No	Default appraisal template for this designation
required_skills	Required Skills	Section Break	No	Layout separator
skills	Skills	Table → Designation Skill	No	Skills required to hold this designation

Print Settings (Frappe Single) — 3 Custom Fields

Field	Label	Type	Reqd	Purpose
compact_item_print	Compact Item Print	Check	No	Single-line item row in PDF (saves paper)
print_uom_after_quantity	Print UOM after Quantity	Check	No	Renders as “5 Box” instead of “Box 5”
print_taxes_with_zero_amount	Print taxes with zero amount	Check	No	Include zero-value tax rows in the print

Address (ERPNext) — 2 Custom Fields

Field	Label	Type	Reqd	Purpose
tax_category	Tax Category	Link → Tax Category	No	Address-level tax category override
is_your_company_address	Is Your Company Address	Check	No	Marks company-owned addresses (used in shipping/branch selection)

Single-Field Custom Fields — 13 DocTypes (1 CF each)

DocType	Field	Label	Type	Purpose
Communication	company	Company	Link → Company	Tag every communication with the company it relates to (for multi-company setups)
Contact	is_billing_contact	Billing Contact	Check	Flag the billing contact for this customer/supplier
Custom DocPerm	impersonate	Impersonate	Check	Allows admin to “log in as” a user with this role
Customer	crm_deal	Frappe CRM Deal	Data	Reference back to Frappe CRM Deal record (when CRM is integrated)
DocPerm	impersonate	Impersonate	Check	Same — standard DocPerm variant

DocType	Field	Label	Type	Purpose
DocShare	impersonate	Impersonate	Check	Same — DocShare variant
Email Account	company	Company	Link → Company	Tag email account with owning company
Income Tax Slab	marginal_relief_limit	Marginal Relief Threshold Limit	Currency	Cap above which marginal relief applies (regulatory threshold)
Project	total_expense_claim	Total Expense Claim (via Expense Claims)	Currency	Roll-up of all Expense Claims against this project
Quotation	crm_deal	Frappe CRM Deal	Data	Reference to CRM deal that produced this quotation
Salary Component	component_type	Component Type	Select (Provident Fund / Additional Provident Fund / ...)	Tags salary components for payroll logic
Task	total_expense_claim	Total Expense Claim (via Expense Claim)	Currency	Same as Project, task-level
Terms and Conditions	hr	HR	Check	Marks T&C template as HR-specific (e.g., appointment letter)
Timesheet	salary_slip	Salary Slip	Link → Salary Slip	Link timesheet to the resulting Salary Slip
UTM Campaign	crm_campaign	Messaging CRM Campaign	Link → Campaign	Bridge field between UTM-tracked campaign and Frappe CRM

Property Setters by DocType

189 Property Setters across **73 DocTypes**. Property distribution:

Property	Count	Meaning
hidden	133	Hide a field/section from the form (still in schema, just not shown)
print_hide	28	Hide from print/PDF (still on form)
default	9	Set the default value for a field
default_print_format	8	Default print format for this DocType
reqd	6	Toggle mandatory on a standard field
mandatory_depends_on	2	Conditional mandatory (reqd if eval condition true)
options	2	Override the options list (for Select fields)
read_only	1	Make a standard field read-only

Sales Invoice (ERPNext) — 20 PSs

- 19 × hidden (mostly section breaks: commission_section, sales_team_section, utm_analytics_section, timesheet_section, etc.)
- 1 × default_print_format = Sales Invoice with Item Image

Sales Order (ERPNext) — 15 PSs

- 14 × hidden (section breaks hidden)
- 1 × default_print_format = Sales Order with Item Image

Delivery Note (ERPNext) — 13 PSs

- 12 × hidden
- 1 × default_print_format = Delivery Note with Item Image

Purchase Invoice (ERPNext) — 11 PSs

- 10 × hidden
- 1 × default_print_format

Quotation (ERPNext) — 10 PSs

- 9 × hidden
- 1 × default = disable_rounded_total=0

Purchase Order (ERPNext) — 10 PSs

- 9 × hidden
- 1 × default_print_format = Purchase Order with Item Image

Purchase Receipt (ERPNext) — 9 PSs

- 9 × hidden

Supplier Quotation (ERPNext) — 8 PSs

- 8 × hidden

POS Invoice (ERPNext) — 6 PSs

- 6 × hidden

Sales Invoice Item / Item / Delivery Note Item / POS Invoice Item / POS Profile (ERPNext) — 17 PSs combined

- Various print_hide for tax_id, hidden for fields not relevant to Haritha

Employee (HRMS) — 4 PSs

- naming_series — reqd=1 (force user to enter a series)
- naming_series — hidden=0 (keep visible)
- employee_number — reqd=0 (make optional since autaname handles it)
- employee_number — hidden=1 (hide from form, since autaname = employee_number)

Shift Type (HRMS) — 2 PSs

- color — default=blue (Tailwind palette default)
- color — options=blue\ncyan\nfuchsia\ngreen\nlime\norange\npink\nred\nviolet\nyellow\n. (Tailwind lowercase palette, was CapitalCase causing Roster crash — see LEARNINGS / Phase 4)

Attendance (HRMS) — 1 PS

- status — options=\nPresent\nAbsent\nOn Leave\nHalf Day\nWork From Home\nHoliday\nWeekly Off (extends the standard 5-value list to 7 for Haritha's roster CSV — see Phase 3.6)

Salary Slip (HRMS) — 2 PSs

- rounded_total — print_hide=0 (show on print)
- 1 × default_print_format

Supplier / Customer (ERPNext) — 4 PSs combined

- naming_series — reqd=0 (autaname can run without prompting)

All other DocTypes — 1 PS each (53 DocTypes)

Mostly hidden on section breaks that Haritha doesn't use (utm_analytics_section, commission_section, sales_team_section, etc.) — single-purpose cleanups that hide unused UI real estate.

Default Print Formats (8 entries)

DocType	Print Format
Sales Invoice	Sales Invoice with Item Image
Sales Order	Sales Order with Item Image
Delivery Note	Delivery Note with Item Image
Purchase Order	Purchase Order with Item Image
(4 more)	(image-branded variants of ERPNext stock formats)

Default Values (9 entries)

DocType	Field	Value
Quotation	disable_rounded_total	0
Sales Order	disable_rounded_total	0
Sales Invoice	disable_rounded_total	0
Delivery Note	disable_rounded_total	0
Purchase Order	disable_rounded_total	0
Purchase Invoice	disable_rounded_total	0
Purchase Receipt	disable_rounded_total	0
POS Invoice	disable_rounded_total	0
Supplier Quotation	disable_rounded_total	0

Reqd (Mandatory) Toggles (6 entries)

DocType	Field	Value	Effect
Employee	naming_series	1	Force series entry (Haritha uses HR-EMP- series)
Employee	employee_number	0	Allow blank (autoname fills it)
Supplier	naming_series	0	Make optional (autoname can default)
Customer (2 more)	naming_series	0	Make optional

Conditional Mandatory (2 entries)

DocType	Field	Condition
Sales Invoice Item	discount_account	(depends_on eval)
Sales Invoice	additional_discount_account	(depends_on eval)

Read-only (1 entry)

DocType	Field	Value	Effect
Packed Item	rate	1	Lock the rate field on packed items (always pull from parent)

Options Override (2 entries — important)

DocType	Field	New options	Why
Attendance	status	Present / Absent / On Leave / Half Day / Work From Home / Holiday / Weekly Off	Extends standard 5-value list to support Haritha's roster CSV (Phase 3.6)
Shift Type	color	Tailwind lowercase palette (blue, cyan, fuchsia, green, lime, orange, pink, red, violet, yellow, ...)	Was CapitalCase, broke Roster Vue SPA. Lowercased to match Tailwind (Phase 4.10)

Print Formats (5 Haritha-affected)

The fixture exports **48 Print Format rows total**. The breakdown:

Module	Count	Notes
HR	2	Haritha-tagged (custom app)
NULL	3	Old stock seeds (LEARNINGS #156 — treat as Haritha-affected)
Accounts, Stock, Selling, Buying, Payroll, Regional	43	Frappe/ERPNext stock formats, included by export

Haritha-affected subset (5)

Name	DocType	Disabled?	Standard?	Type	Purpose
Job Offer	Job Offer	No	Yes	Jinja	HR module standard offer letter format
Standard Appointment Letter	Appointment Letter	No	Yes	Jinja	HR module standard appointment letter
Drop Shipping Format	Purchase Order	No	Yes	Jinja	Stock format, dropped from POs not for stock

Name	DocType	Disabled?	Standard?	Type	Purpose
Cheque Printing Format	Journal Entry	No	Yes	Jinja	Stock format, renders cheque layout
Payment Receipt Voucher	Journal Entry	No	Yes	Jinja	Stock format, prints payment voucher

Why are these here? All 5 export cleanly via `{"doctype": "Print Format", "filters": {"module": "haritha_hospital"}}` because: - 2 are actually `module=HR` (truly Haritha-owned after the export filter) - 3 are `module=NULL` and slip through any `module=...` filter — they're stock seeds Frappe never back-filled (LEARNINGS #156). Decision rule applied: included because they're functionally relevant to Haritha operations.

Notifications (6 Haritha-affected)

The fixture exports **8 Notification rows total**.

Module	Count	Notes
HR	3	Haritha-tagged Stock notifications (Fiscal Year, Retention Bonus, Material Request)
Accounts, Payroll, Manufacturing	3	
NULL	2	Old stock seeds (Error Log, Integration Request)

Haritha-affected subset (6)

Name	Document Type	Event	Enabled	Channel	Purpose
Training Scheduled	Training Event	Submit	Yes	Email	Notifies employees when they're added to a training event
Training Feedback	Training Result	Submit	Yes	Email	
Exit Interview Scheduled	Exit Interview	Days Before	Yes	Email	Notifies manager when training feedback is submitted Reminder to employee N days before scheduled exit interview

Name	Document Type	Event	Enabled	Channel	Purpose
Notification for new fiscal year	Fiscal Year	New	Yes	Email	Stock — admin alert when FY is created
Retention Bonus	Retention Bonus	Days Before	Yes	Email	Stock — bonus due reminder
Material Request Receipt Notification	Material Request	Value Change	Yes	Email	Stock — alert when MR status changes

Note: TRACKER and HARITHA_HOSPITALS_GUIDE.md §5.4 say “2 notifications” — that was the count as of an earlier export. The 2026-08-29 export now shows 3 HR-module Notifications (Training Scheduled + Training Feedback + Exit Interview Scheduled). All 3 are HRMS stock notifications re-tagged as Haritha when first set up — they’re functionally Haritha-relevant. The fixtures export includes 3 stock Notifications (Accounts, Payroll, Manufacturing modules) and 2 NULL-module old stock seeds (Error Log, Integration Request), for a fixture total of 8.

Letter Heads (2)

Letter Head DocType has **no module column** (LEARNINGS #155), so the fixtures export includes ALL rows:

Name	Letter Head Name	Is Default	Disabled	Purpose
Company Letterhead	Company Letterhead	No	No	First letterhead — generic company branding
Company Letterhead - Grey	Company Letterhead - Grey	Yes	No	Default letterhead — grey-themed variant used for Haritha prints

Both are HRMS / ERPNext stock seeds; neither is Haritha-customized in content. They live in the fixtures file because (a) the export filter is "Letter Head" without a module filter, and (b) they’re the only Letter Heads on the system, so removing them would break print branding.

How customizations are deployed

The “module = haritha_hospital” tag

Frappe v16 uses a module column on Custom Field, Property Setter, Print Format, and Notification (LEARNINGS #152). The hooks.py filter selects rows WHERE module='haritha_hospital'.

Diagram

Figure 1: Diagram

Without this tag, `bench export-fixtures` exits 0 silently doing nothing (LEARNINGS #151).

If you create a Custom Field via the Frappe UI without setting `module='haritha_hospital'`, it WILL be saved (module defaults to the parent app, e.g., HRMS), but **it will NOT export** with the Haritha fixture. Always manually re-tag the module via UI or direct DB UPDATE after creation.

How to add a new customization

Recommended pattern:

1. **Plan** — which DocType? which field? what type? required? hidden? default? Print Format template? Notification trigger?
 2. **Tag it** — set `module='haritha_hospital'` on the new row (UI or DB).
 3. **Test on dev first** — never on prod. Make the change on `pberpdev.duckdns.org`, validate, then mirror to prod.
 4. **Export fixtures:**

```
docker exec erp-prod-backend-1 bash -c \  
  "cd /home/frappe/frappe-bench && \  
  bench --site pberpprod.duckdns.org export-fixtures --app haritha_hospital"
```
 5. **Verify the JSON:**

```
git diff apps/haritha_hospital/haritha_hospital/fixtures/*.json
```

Confirm the new row is there and `module='haritha_hospital'`.
 6. **Commit + push:**

```
git add apps/haritha_hospital/haritha_hospital/fixtures/  
git commit -m "feat(custom): add <field> to <DocType>"  
git push origin main
```
 7. **Apply on other envs:**

```
docker exec erp-<env>-backend-1 bash -c \  
  "cd /home/frappe/frappe-bench && \  
  bench --site <env>.duckdns.org migrate"
```

(migrate re-runs fixtures; idempotent.)
 8. **Restart backend + workers** — `gunicorn --preload caches sys.path` (LEARNINGS #153):

```
docker restart erp-<env>-backend-1 erp-<env>-scheduler-1 \  
  erp-<env>-queue-short-1 erp-<env>-queue-long-1
```
-

Known limitations

Limitation	Detail	Workaround
Letter Head has no module column	The export filter "Letter Head" cannot include a filters dict — exports ALL rows.	Acceptable — only 2 LHs in the system. If Haritha adds custom LHs later, they'll come along too.
Print Format / Notification with module=NULL	Old stock seeds never back-filled by Frappe. They slip through the module=... filter.	Decision rule (LEARNINGS #156): if creation_date < 2024 AND module IS NULL, treat as stock. Documented in this doc.
Custom Field module defaults to parent DocType's app	UI-created Custom Fields get module='HRMS' (or whichever app owns the DocType), not 'haritha_hospital'.	Manually re-tag in UI or via <code>frappe.db.sql("UPDATE tabCustom Field SET module='haritha_hospital' WHERE name=...")</code> after creation.
HRMS core can't be modified	We can't fix hardcoded controller-level checks (e.g., HRMS Attendance.validate() 5-value status list — LEARNINGS #105) by Property Setter alone.	Monkey-patch in scripts (bulk_submit.py pattern) OR live with the limitation.
Frappe Property Setter for options doesn't refresh bench console cache	After setting Select options, need a NEW bench console session to see new options. Within same session, meta is cached (LEARNINGS, post-Phase 3.6).	Restart workers or open a fresh bench console session.
Stock Notifications/Print Formats get exported too	Any stock row that survives the filter gets included.	Acceptable — they're small (~5 rows), and removing them would mean hand-curating fixtures.

End of 01.1. Next: 01.2 — Schema Diagram (ERD).